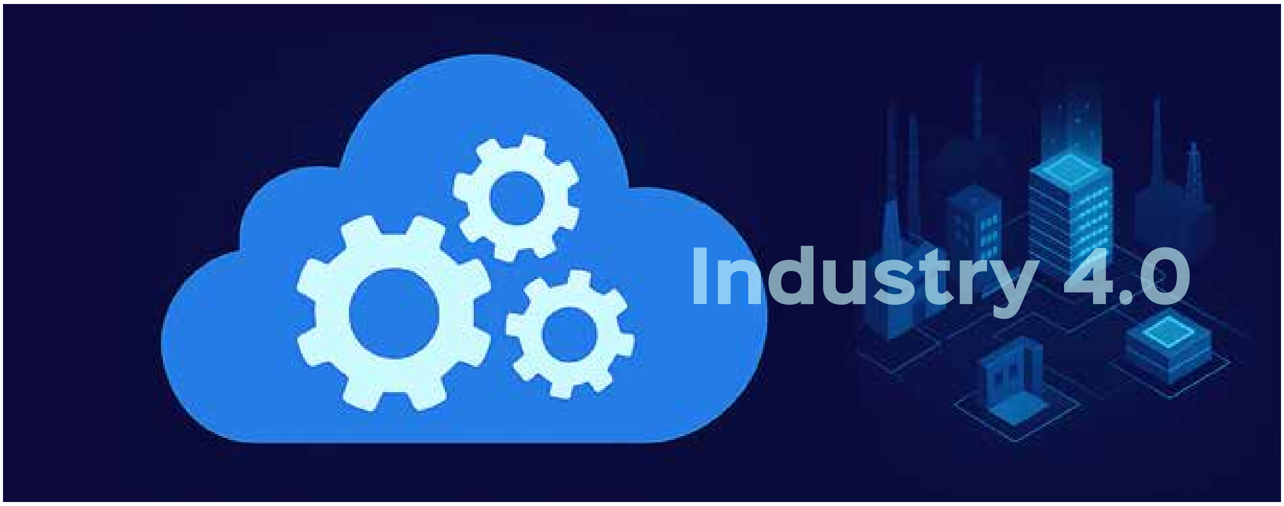
Industrial Clouds: Meeting the Needs of Industry 4.0

Microsoft, Amazon and Google have introduced industrial clouds to fulfil the demands of Industry 4.0. These cloud services meet the special requirements of industrial companies in the AI age.



Industry 4.0, the fourth industrial revolution, is reshaping the way organisations design, manufacture, operate, and service their products and processes. To thrive in this new era, industrial organisations need to leverage the power of cloud computing, artificial intelligence (AI), and the Internet of Things (IoT) to optimise their operations, enhance their productivity, and innovate faster.

However, the industrial sector also faces unique challenges and has requirements that demand a specialised and tailored approach to cloud computing. These include:

- The need to integrate and manage heterogeneous and legacy systems, devices, and data sources across different locations and environments, such as factories, warehouses, offices, and field sites.
- The need to ensure high levels of security, privacy, compliance, and reliability for sensitive and mission-critical industrial data and applications, especially in regulated and complex industries, such as manufacturing, energy, healthcare, and transportation.
- The need to address the skills gap and talent shortage in the industrial workforce, as well as to empower and enable workers with the right tools and information to perform their tasks more efficiently and effectively.
- The need to deliver scalable, flexible, and cost-effective solutions that can adapt to the changing and dynamic needs and demands of the industrial market and customers.

Industrial clouds

To address these requirements, Microsoft, Google and Amazon have developed the industrial cloud (Table 1), which enables industrial organisations to:

- Build, deploy, and manage large-scale and complex industrial solutions using various methods and modes of training, such as supervised, unsupervised, semi-supervised, and reinforcement learning, as well as various levels and degrees of automation, such as manual, automated, and hybrid.
- Provide and ensure high levels of accuracy and precision for industrial solutions, mainly focused on data and model management, using various types of data and information, such as text, speech, image, video, and audio.
- Simplify and unify the development and deployment of natural language processing (NLP) technologies and services.

Cost benefits of using the industrial cloud

Industrial cloud solutions offer a range of cost benefits that can significantly impact the financial health and